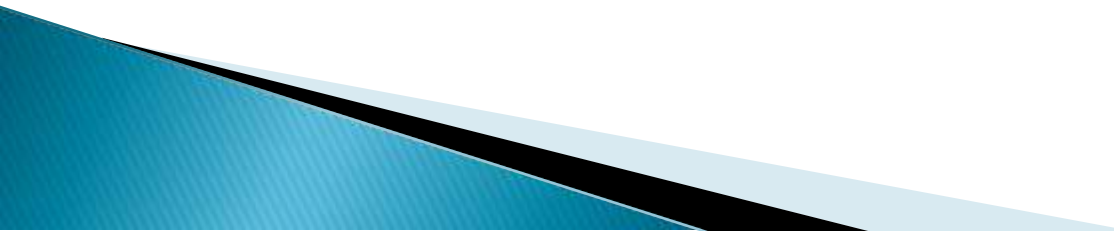


Common diseases of teeth and supporting structures

RADIOGRAPHIC APPEARANCE OF COMMON Inflammatory DENTAL & jaw DISEASES



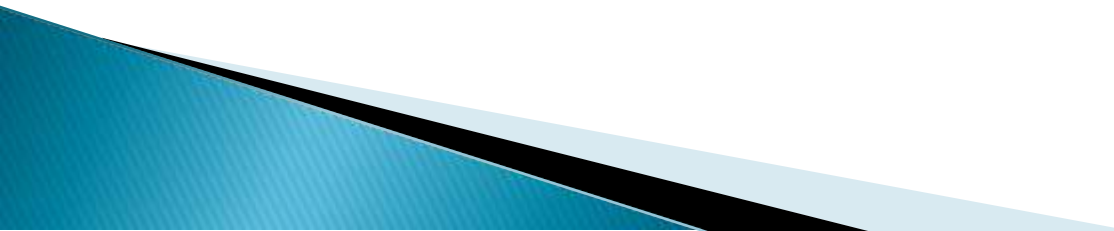
COMMON DENTAL DISEASES

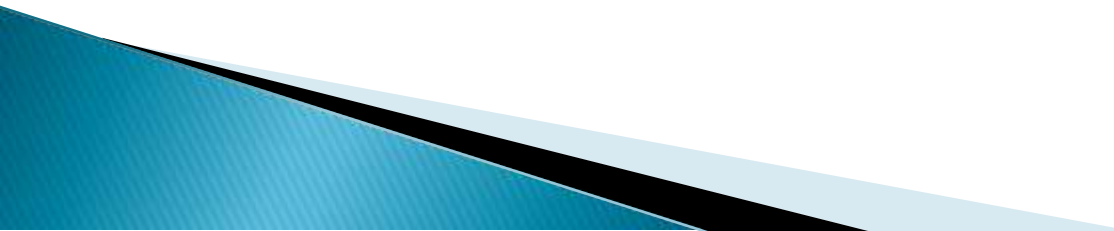
- ▶ Dental caries
 - ▶ Periodontal Diseases
 - ▶ pericoronitis
 - ▶ Inflammatory lesions of the jaws
 - ▶ Osteomyelitis.
- 

dental caries

Dental caries, the most common disease in the mouth, is also the most common disease of the entire body, affecting **95%** of population.

Radiography is useful for detecting dental caries because the carious process causes tooth **demineralization**.



- ▶ The carious lesion (**the demineralized area of the tooth that allows greater infiltration of x-rays**) is darker (more radiolucent) than the unaffected portion and may be detected on radiographs.
 - ▶ Intraoral radiography can reveal carious lesions that otherwise might go **undetected during clinical examination.**
- 

- ▶ A clinical examination may be able to identify carious lesions on the occlusal and exposed smooth surfaces of the teeth. However, it is nearly impossible to clinically identify caries occurring on the proximal surfaces of teeth (interproximal caries) unless there has been cavitation.

Factors affecting the interpretation of dental caries

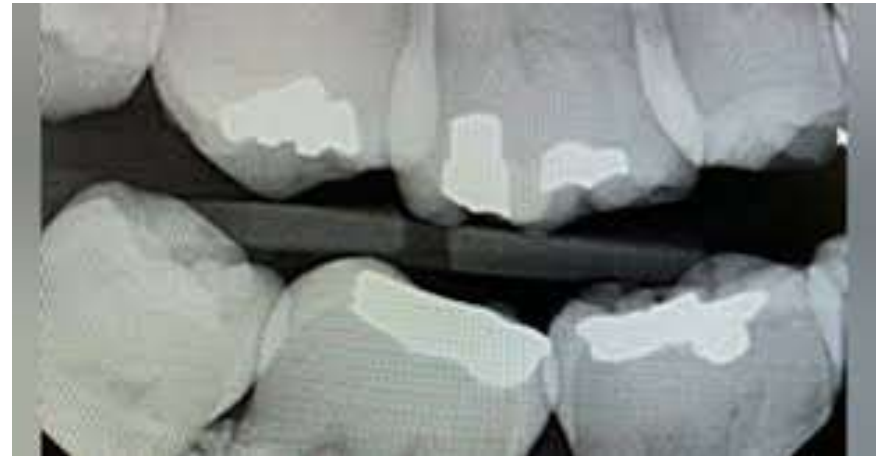
1. **peripheral burnout** It could be reduced by decreasing exposure time or Kv



2. Ratio of enamel to caries which X-ray photon must penetrate.
3. Errors in vertical angulations: Increased & decreased.



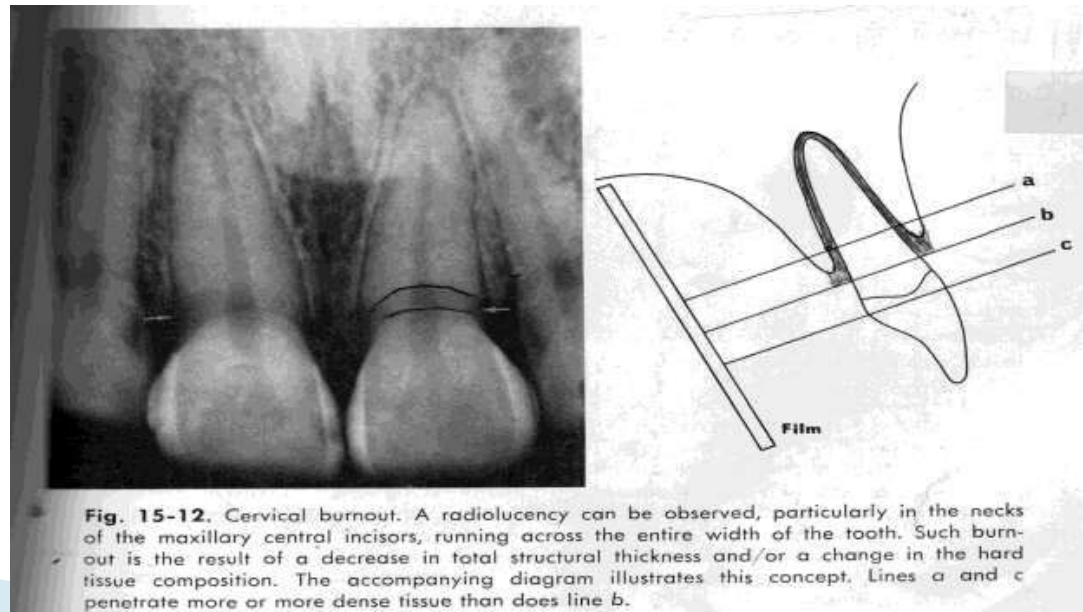
4.errors in horizontal angulations. which cause interproximal overlapping



5. Lesions simulate dental caries:
example developmental defects

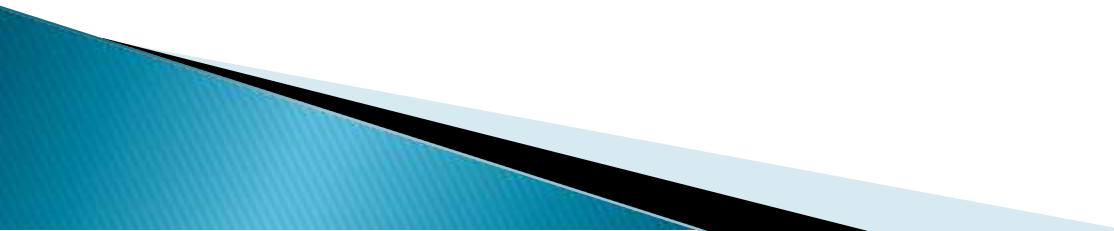


6. Cervical burnout: Radiolucent band around the neck of the teeth. It can be reduced by reducing kvp

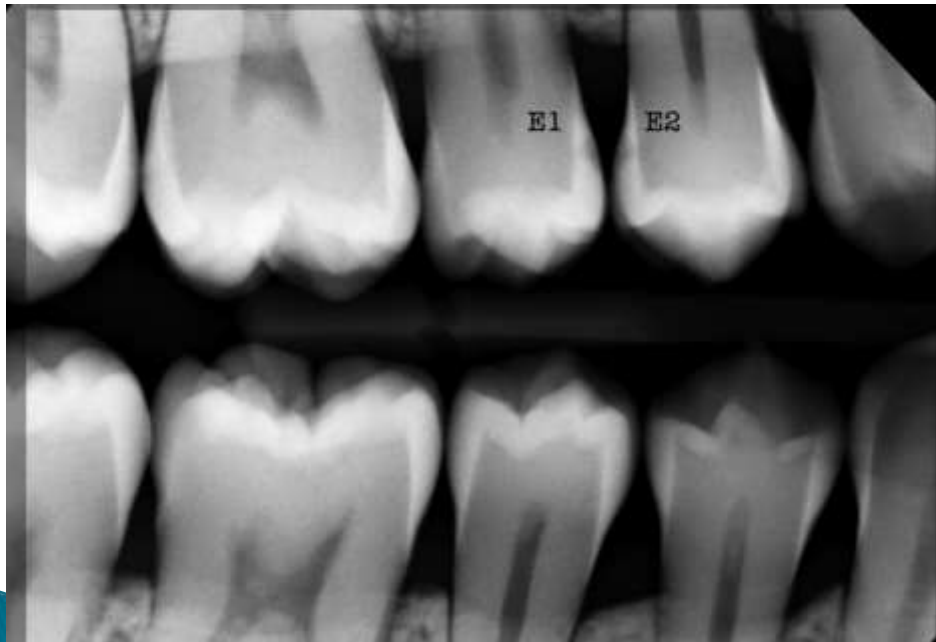


Interproximal caries

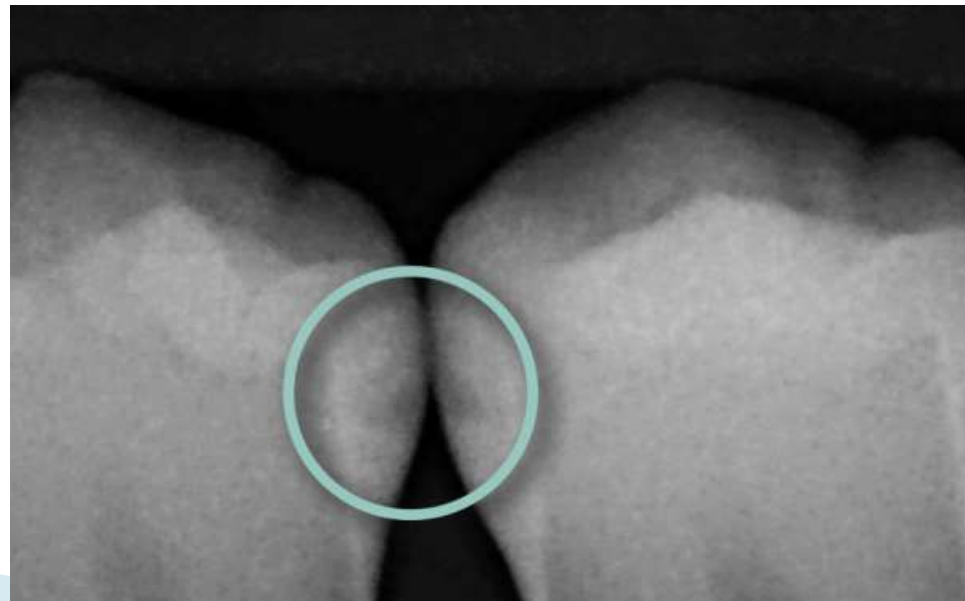
The intra oral radiographic film, notably the **bitewing** film or the radiogram made with the **paralleling technique** is extremely **useful** in detecting **interproximal carious lesions**.



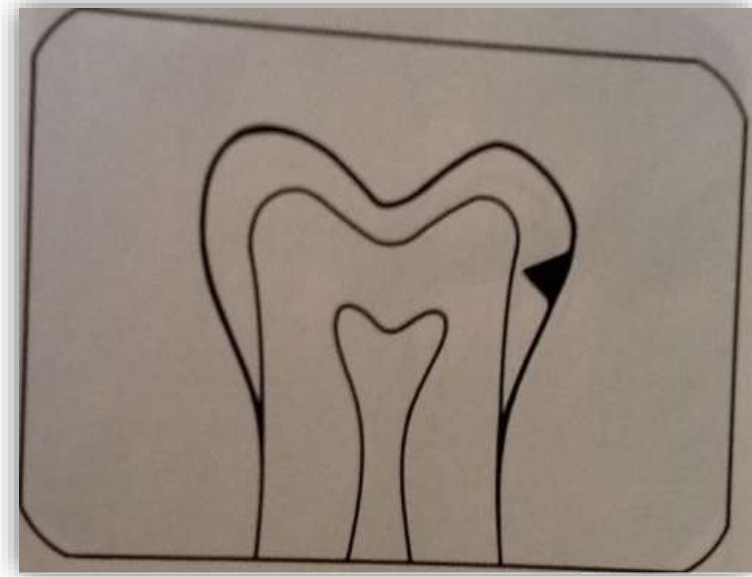
The first evidence of the interproximal carious lesion consists of an extremely **small notching** in the enamel surface **below** the interproximal contact point.



An incipient interproximal lesion extends **less than halfway** through the thickness of enamel. The term incipient means beginning to exist or appear. An incipient lesion is seen in **enamel only**.



- ▶ A moderate interproximal lesion extends **greater than halfway** through the thickness of enamel, but does not involve the DEJ. A moderate lesion is seen in **enamel only**.

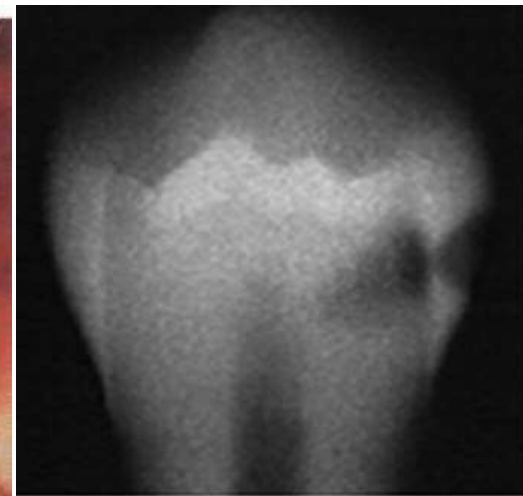
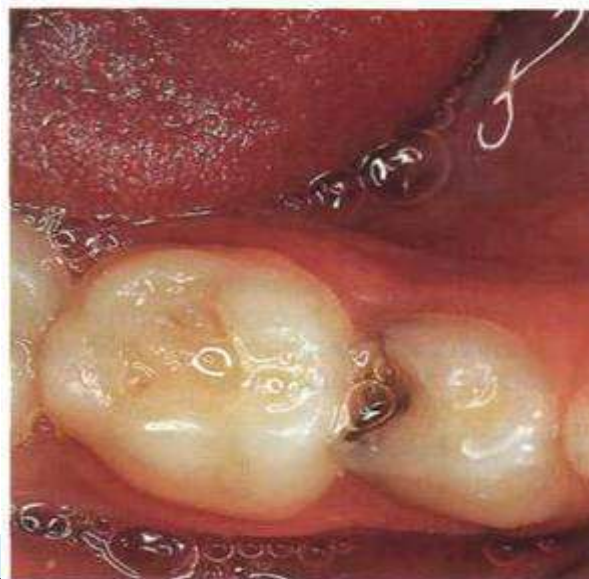


An advanced interproximal lesion extends to the DEJ or through the DEJ An advanced and into the dentin, but does not extend through the dentin greater than half the distance toward the pulp. An advance lesion affects both enamel and dentin.



- ▶ A severe interproximal lesion extends through enamel, through the **dentin and greater than half** the distance towards the pulp. A severe lesion involves both the **enamel and dentin** and may clinically appear as cavitation(or hole) in the tooth. Avoid →

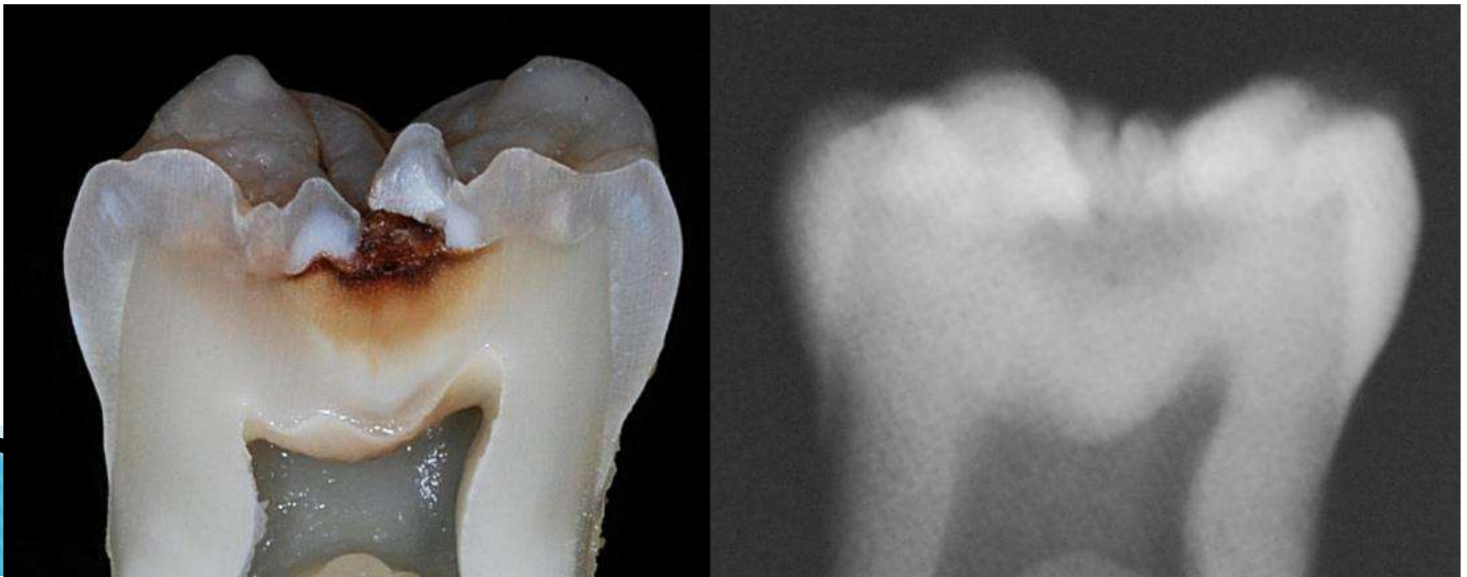
- ▶ Horizontal overlap



Case 5

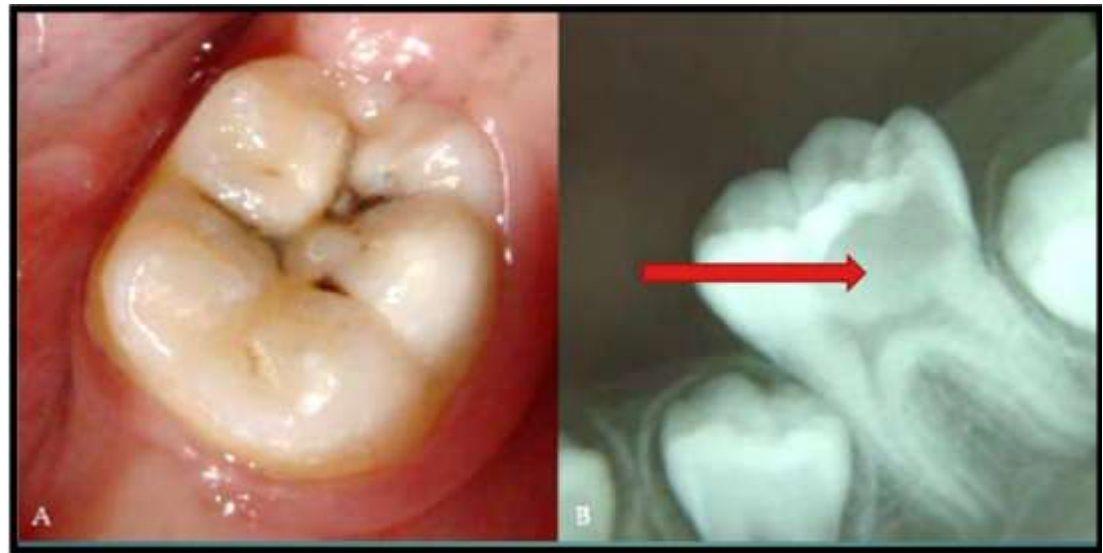
occlusal caries

- ▶ The first radiographic sign is a **dark line between enamel and dentin** occasionally occlusal caries is **confused** with **buccal of lingual caries** and differentiated clinically.



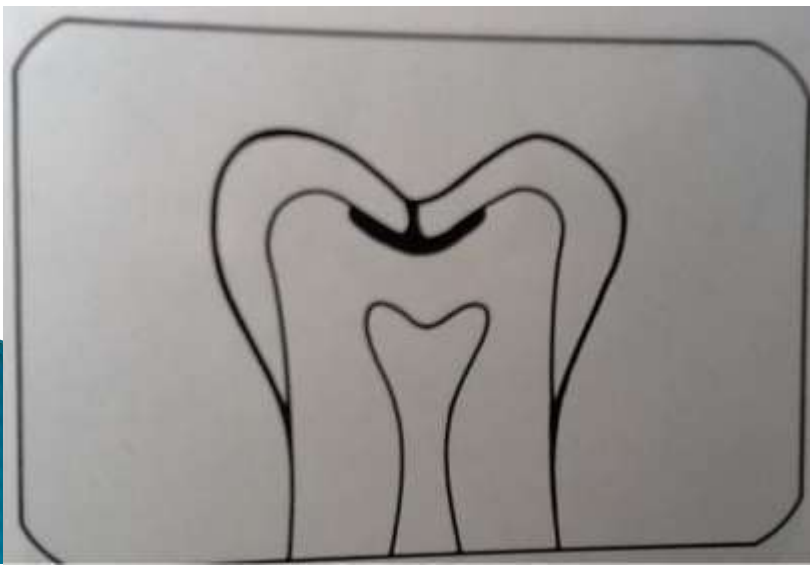
Incipient occlusal caries

Cannot be seen on a dental radiograph and must be detected clinically.



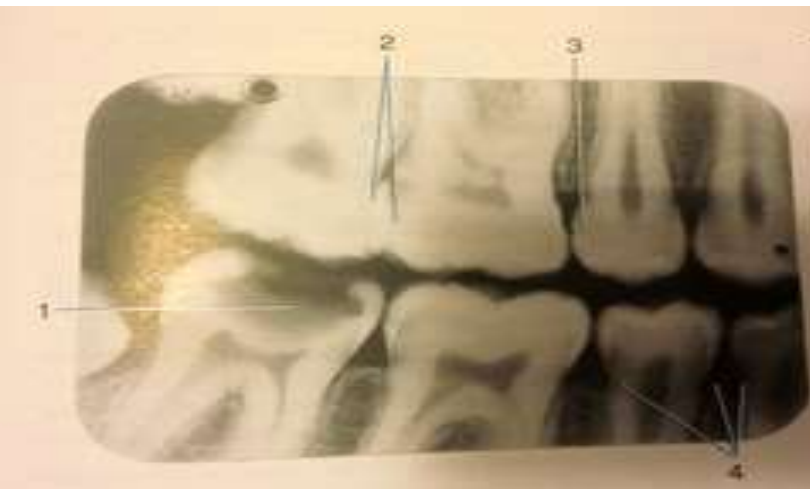
Moderate occlusal caries

- ▶ extends into dentin and is seen as a **very thin radiolucent line**.
- ▶ The radiolucency is located under the enamel of the occlusal surface of the tooth



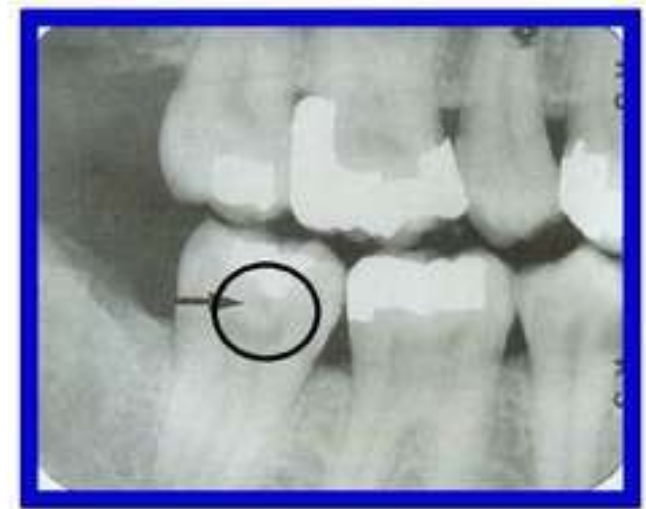
Severe occlusal caries

- ▶ extends into dentin and is seen as a large radiolucency.
- ▶ Severe occlusal caries is apparent clinically and appears as a cavitation(or hole) in a tooth.



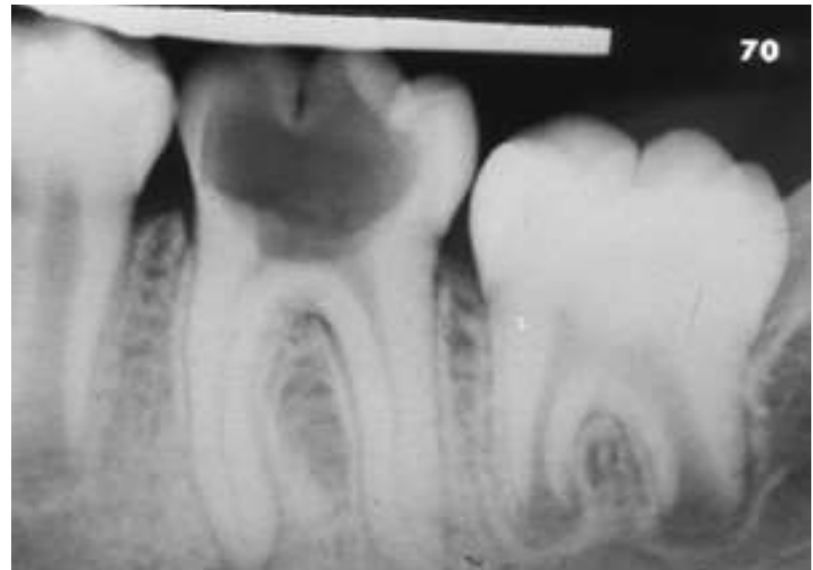
Buccal and lingual caries

It occurs in the pits and grooves; it is **elliptical and / or semi lunar**. It is **difficult to differentiate** between buccal and lingual caries, also it may be **confused with pulp exposure** even the lesion may be relatively superficial.



pulp exposure

Radiographic evidence suggesting pulp exposure should not be used as the only definitive criterion for either tooth removal or endodontic therapy?????.



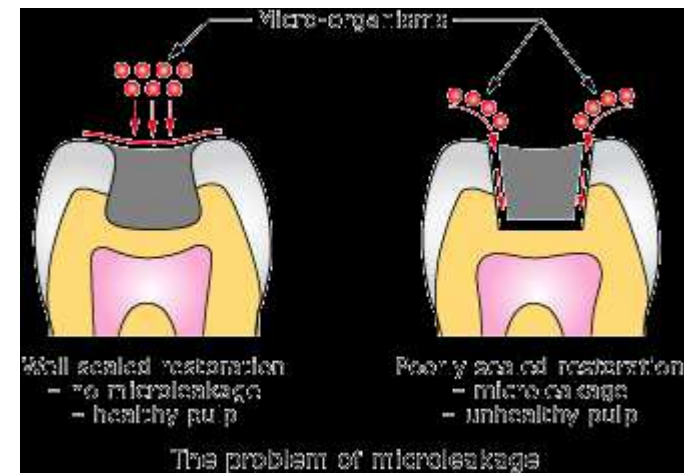
Cemental caries

- ▶ On a dental radiograph, root surface caries appears as a **cupped-out or crater-shaped** of varying depth radiolucency just below the cementoenamel junction .
- ▶ **Early lesions may be difficult to detect on dental radiograph**



Secondary or recurrent caries

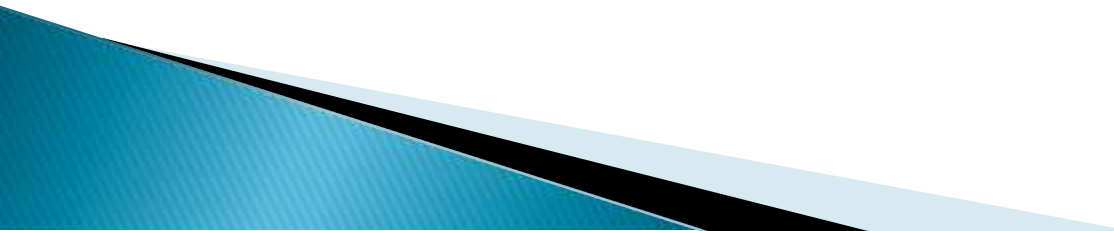
- ▶ occurs adjacent to a **pre- existing restoration**.
- ▶ Caries occurs in this region **because** of inadequate cavity preparation, defective margins or incomplete removal of caries prior to the placement of the restoration



- ▶ On a dental radiograph, recurrent caries appears as a radiolucent area just **beneath a restoration**. Recurrent caries is most often seen **beneath the interproximal margins of a restoration**

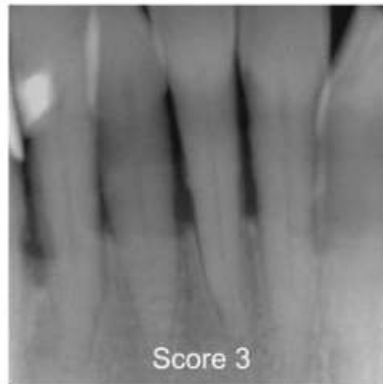
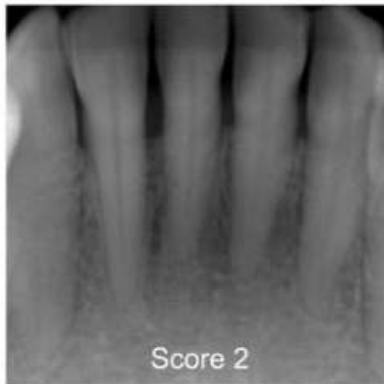
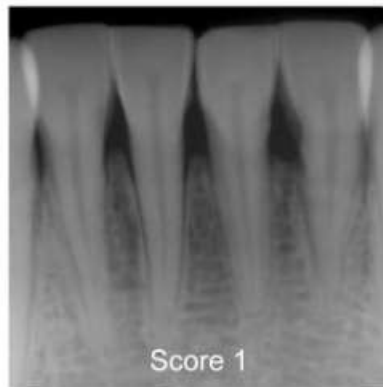
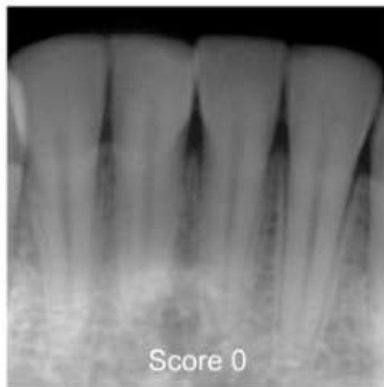


Periodontal Diseases

- ▶ The most common of periodontal disease are **gingivitis** and **periodontitis**.
 - ▶ The clinical examination should include **periodontal probing**, a **gingival index**, mobility charting, and an evaluation of the amount of attached gingiva.
- 

Radiographic features of healthy periodontium

- ▶ relationship between the crestal bone margin and the cemento–enamel junction (CEJ).
- ▶ this distance is within normal limits (**2–3 mm**)



Radiographic features of periodontal disease

- ▶ Acute and chronic gingivitis Radiographs provide no direct evidence of the soft tissue involvement in gingivitis.
- ▶ Periodontitis is the name given to periodontal disease when the superficial inflammation in the gingival tissues extends into the underlying alveolar bone and there has been loss of attachment

- ▶ The destruction of the bone can be either **localized** affecting a few areas of the mouth, or **generalized** affecting all areas.

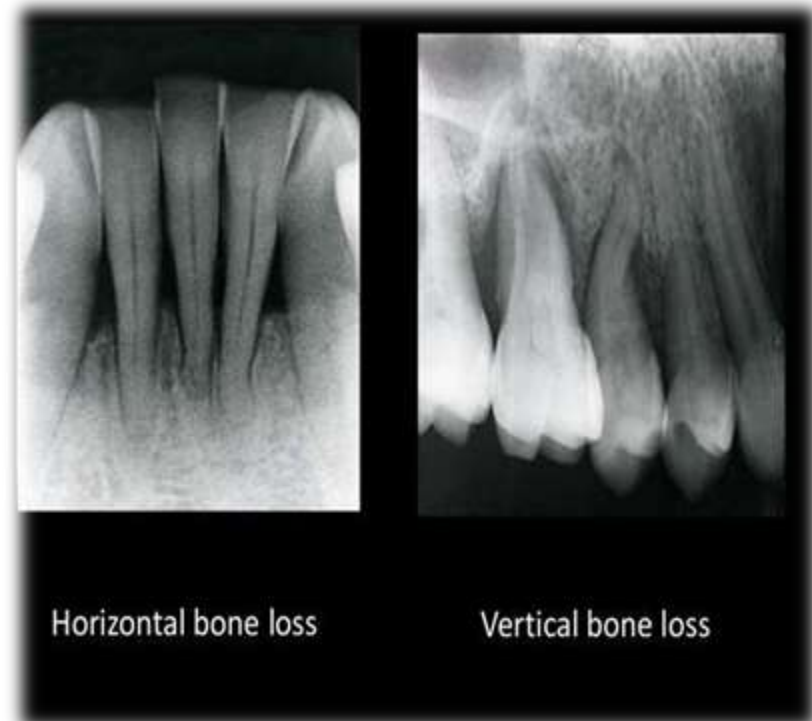


The terms used to describe the various appearances of bone destruction include:

- ▶ Horizontal bone loss
- ▶ Vertical bone loss
- ▶ Furcation involvements.

The amount of bone loss

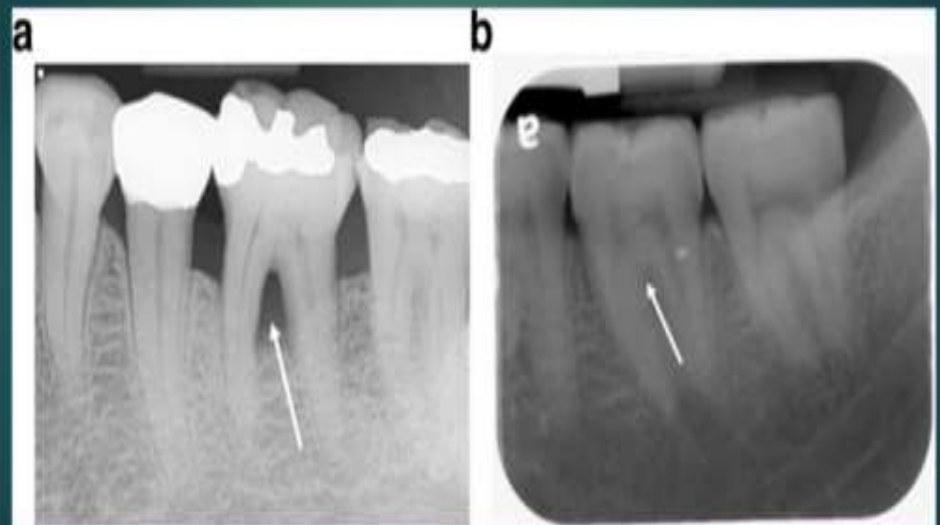
mild,
moderate
or severe.



Vertical bone resorption



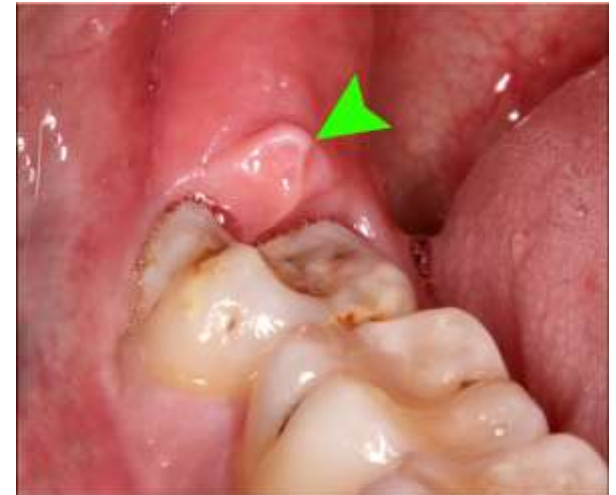
Furcation involvement



Horizontal bone loss

PERICORONITIS

- ▶ As **mandibular third molar** is the most commonly impacted tooth and its position is in such a way that its pericoronal flap gets frequently traumatised, it is the most commonly affected region.
- ▶ Patient usually presents with **swelling** in affected area and **inability to open the mouth** completely. Patient may be having **severe pain**.



Radiographic Features

- ▶ The most common radiographic feature of pericoronitis of mandibular third molar is that there is **presence of distal bone loss**.
- ▶ This distal bone loss is **semilunar or circumferential** in shape.
- ▶ **In the case of mesially tilted impaction, bone loss is present on the mesial side.**

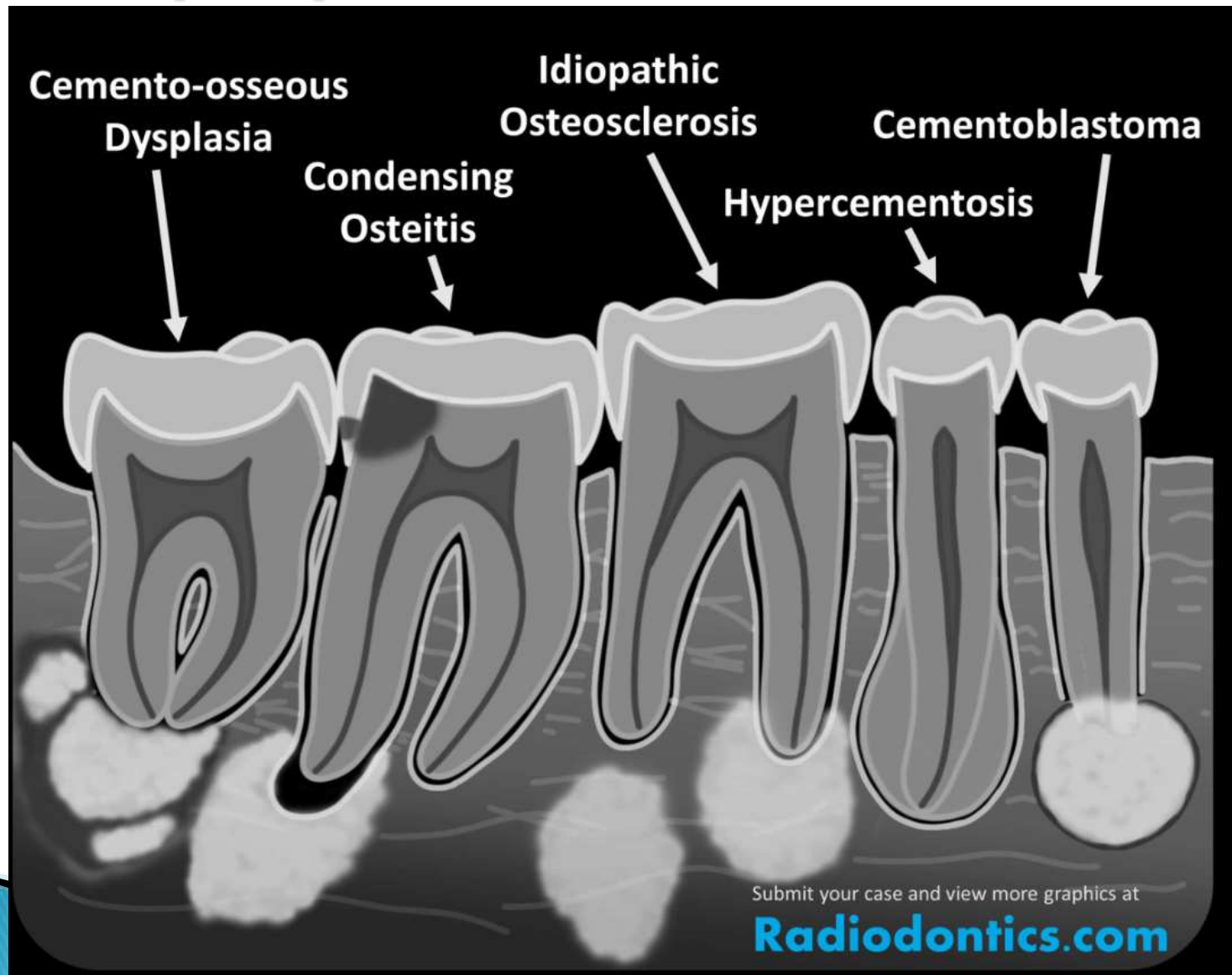


Inflammatory lesions of the jaws



FIG. 22.1 The development of periapical inflammatory disease at the tooth root apex and dissemination into the adjacent bone.

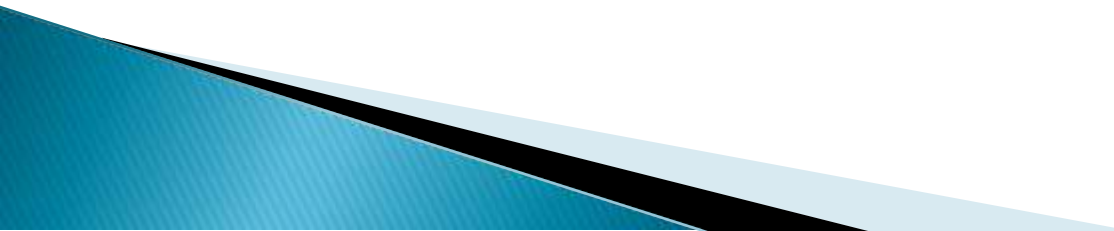
Radiopaque



Radiolucent



1. Granuloma Periapical Radiograph Image

- ❖ Differences between **cyst and granuloma**: It is accepted that lesions having a diameter smaller than 1.5 cm are apical granulomas and lesions bigger than this are periapical cysts
 - ❖ Difference between **granuloma and apical abscess**: Apical abscesses form large radiolucencies with diffuse irregular borders.
 - ❖ **Periapical scar**: Periapical scar is associated with endodontically treated tooth
 - ❖ Periapical osteofibrosis → vital tooth
- 

Periapical radiolucencies

Interpretation of periapical radiolucency

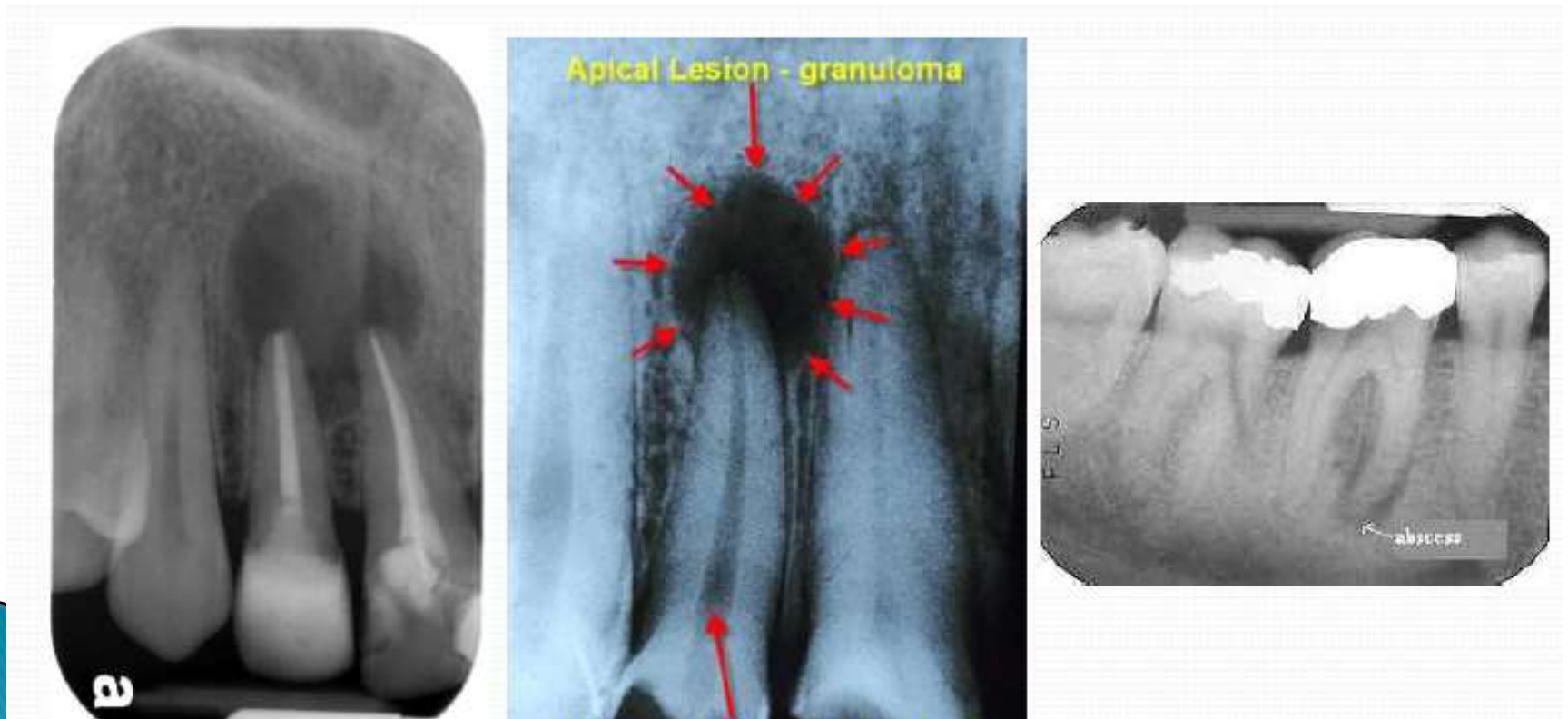
1. Superimposition of radiolucent normal anatomical structure like mental foramen & maxillary sinus, & inferior dental canal.



2. Incomplete root formation (with children)



3. Periapical cyst , granuloma, or abscess



4. First stage of cementoblastoma



Root end changes

- ▶ 1. Hypercementosis: clubbing of root (radiopacity related to the root surface)



2. Root resorption

A. smooth root resorption(smooth root periphery).

B. Rough root resorption (roughened root periphery).



Bone changes associated with apical alteration

- ▶ 1. Condensing osteitis: bone sclerosis as a result of stress, trauma or infection, it characterized by reduction of size of trabecular spaces.



2-Ankylosis: bone and tooth roots occasionally become fused.

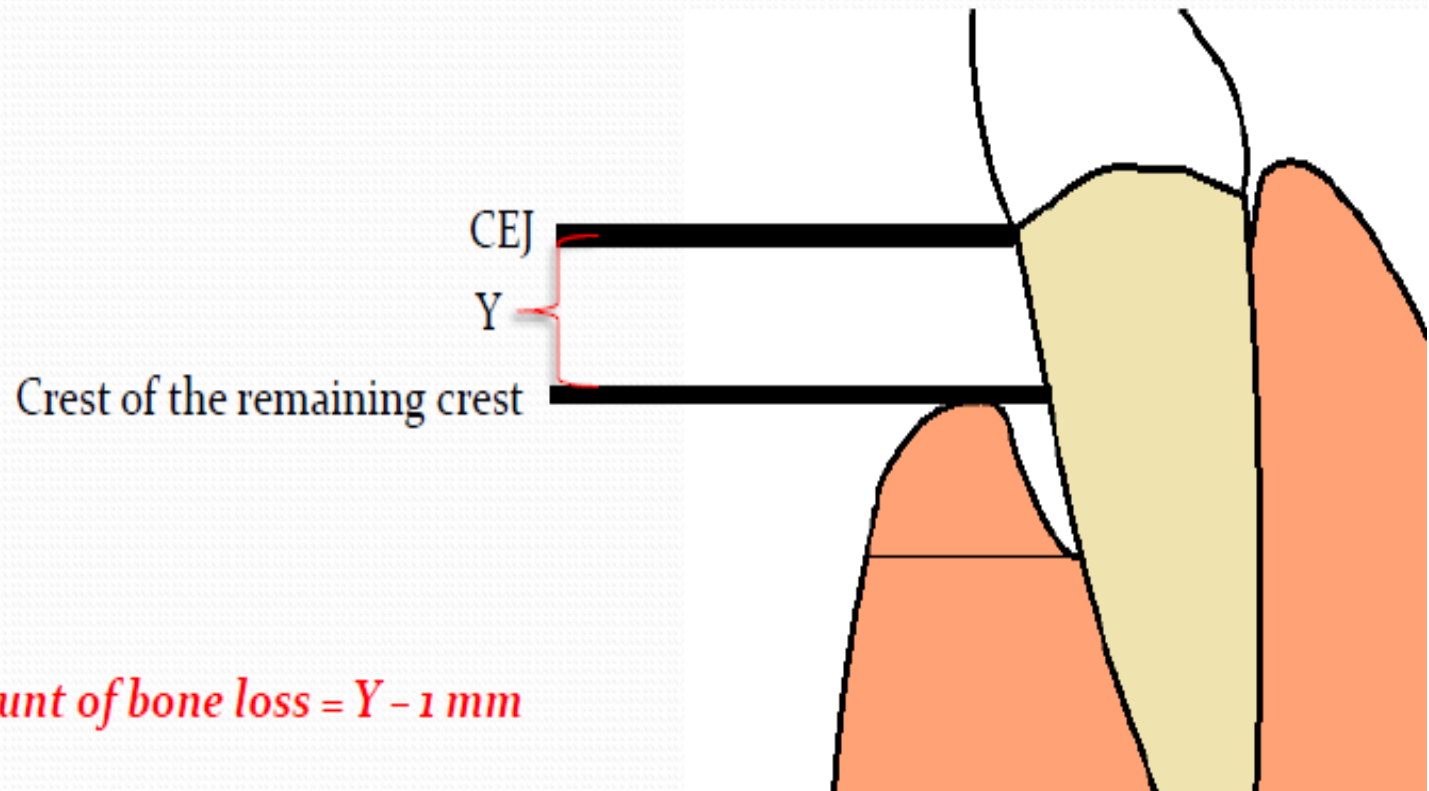


location of bone loss

- ▶ periodontal bone loss may be restricted to one or a few areas.
- ▶ Generalized periodontitis: when periodontal bone loss is evenly distributed throughout the mouth.



Amount of bone loss



Direction of bone loss

1. Horizontal bone loss:-when loss occurs on a plane that is parallel with a line drawn.
2. Vertical bone loss:- when there is greater bone loss in one tooth than on the adjacent tooth, so the bone level is not parallel with a line joining the cementoenamel junctions





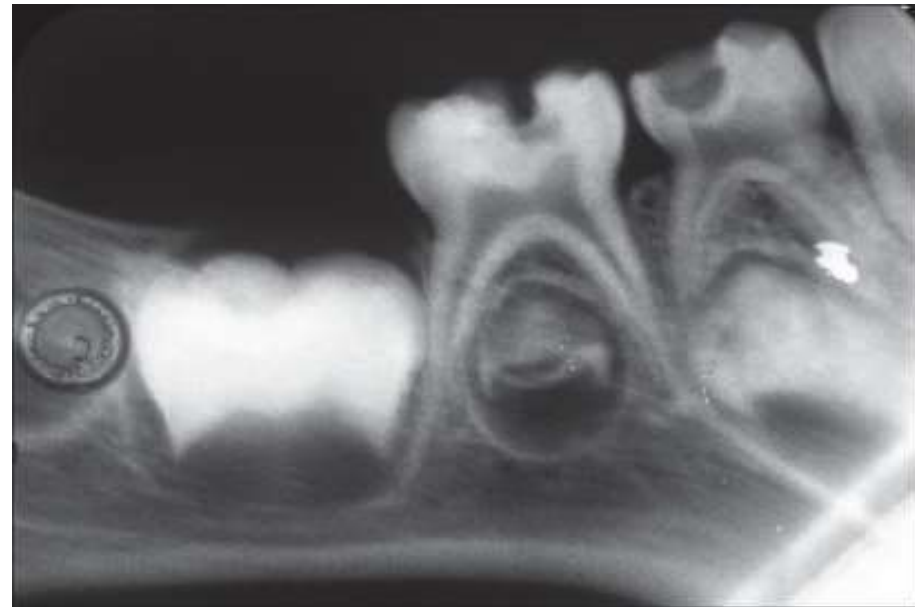
Horizontal bone loss



Vertical bone loss

Tooth resorption:

- ▶ 1. Physiologic root resorption: resorption of the deciduous teeth normally precedes their exfoliation



- ▶ 2. Idiopathic tooth resorption: resorption of tooth surfaces, either internal or external can occur from unknown cause



- ▶ 3. Pathologic tooth resorption: -it usually caused by pressure, infection , neoplasm or trauma



(From Ileen O. Phelan J. Oral pathology for the dental hygienist, ed 4, St Louis, 2004, WB Saunders)

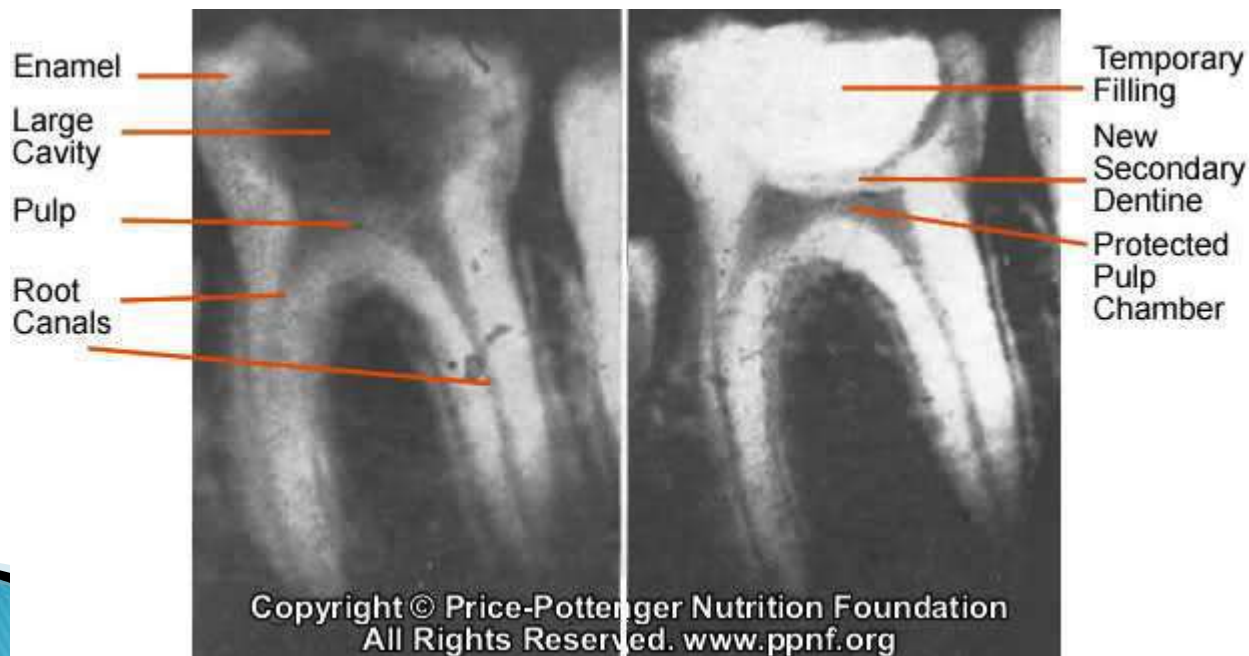
Pulp calcifications

It include pulp stones secondary dentin dentinal bridges and pulpal obliteration

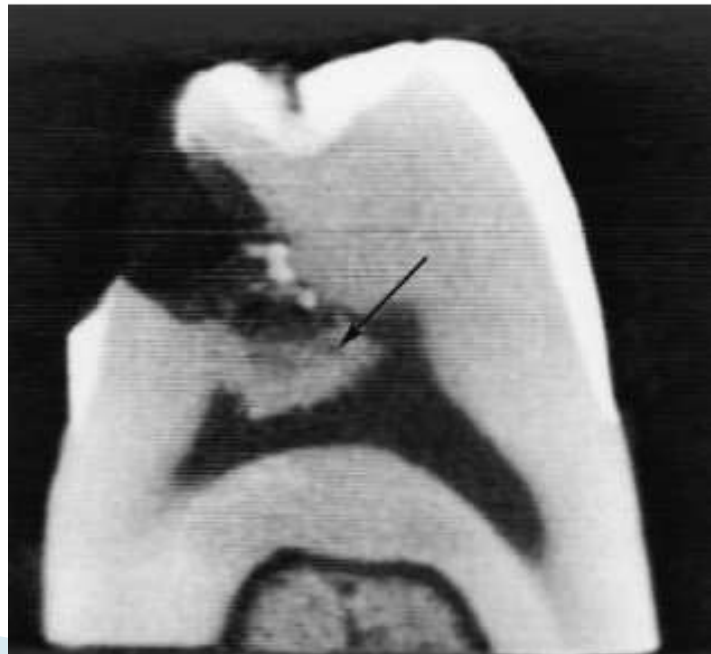
1. Pulp stone:- round or oval opacities within the pulp



- ▶ 2. Secondary dentin:- reduces the size of the chamber. It appears to be a normal aging phenomenon as well as a defense mechanism.



- ▶ 3. Dentinal bridges:- develop between Normal pulp tissue and large carious lesion



- ▶ 4. Pulpal obliteration: is associated with aging.



Tooth fractures

- ▶ Fracture-line and discontinuity in the outline of the tooth are the most usually observed signs of fracture.



Erosion, abrasion and attrition

- ▶ Loss of tooth structure can be observed radiographically

